

Exploratory Data Analysis (EDA)

Purpose:

The purpose of this exploratory data analysis (EDA) is to describe patterns in EFN service reach, utilization, and food distribution capacity over time and across service units. The analysis examines aggregate trends, repeat utilization behavior, demographic composition, and cross-unit heterogeneity to characterize service activity within the dataset.

This section is descriptive in nature and provides a structured overview of service activity within the observed EFN network. No causal claims, policy evaluations, or performance assessments are made.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

```
In [2]: df = pd.read_excel("2014-2024_meals_no_partners.xlsx")
```

EDA Structure

This exploratory analysis proceeds in the following order:

1. Dataset overview and coverage
2. Service reach over time
3. Service intensity (repeat utilization)
4. Food distribution capacity
5. Demographic composition
6. Variation across service units
7. Summary of descriptive patterns

All results are descriptive in nature and do not imply causal relationships.

1. Dataset Overview and Coverage

Numbers of Rows and Columns

```
In [3]: # Overall dataset dimensions
num_rows, num_cols = df.shape

num_rows, num_cols
```

Out[3]: (348, 22)

Time Coverage

```
In [4]: # Time coverage
year_min = df["year"].min()
year_max = df["year"].max()
num_years = df["year"].nunique()

year_min, year_max, num_years
```

Out[4]: (2014, 2024, 11)

Service Unit Coverage

```
In [5]: # Service unit coverage coverage
num_service_units = df["zip"].nunique()

num_service_units
```

Out[5]: 36

Observations Per Year

```
In [6]: # Observations per year
obs_per_year = (
    df.groupby("year")
      .size()
      .reset_index(name="num_service_units")
)

obs_per_year
```

```
Out[6]:
```

	year	num_service_units
0	2014	29
1	2015	29
2	2016	29
3	2017	30
4	2018	30
5	2019	31
6	2020	31
7	2021	33
8	2022	36
9	2023	36
10	2024	34

Observations Per Service Unit

```
In [7]: # Observations per service unit
obs_per_service_unit= (
    df.groupby("zip")
      .size()
      .reset_index(name="num_years_observed")
)

obs_per_service_unit.describe().round(1)
```

```
Out[7]:
```

	num_years_observed
count	36.0
mean	9.6
std	2.9
min	2.0
25%	10.8
50%	11.0
75%	11.0
max	11.0

The dataset spans calendar years from 2014 to 2024 and contains 348 service unit–year observations across 36 distinct service units. Coverage varies by year

and service unit, consistent with an unbalanced panel structure driven by changes in partner participation, service coverage, and reporting practices.

Most service units exhibit stable longitudinal coverage. The median service unit appears in all observed years, and a majority of service units are present for at least ten years. A small number of service units enter or exit the dataset earlier, consistent with changes in partner participation, program availability, or reporting practices. Overall, the panel is nearly balanced, with a stable core of service units observed throughout the sample period.

These characteristics are taken into account in subsequent descriptive analyses.

2. Service Reach Over Time

This section examines how the number of individuals served by EFN changes over time, using total and unduplicated client counts. All patterns are descriptive.

Service reach is measured as the number of individuals receiving food assistance within a calendar year. Unduplicated client counts reflect the breadth of reach (unique individuals served), while total client counts reflect overall service volume.

```
In [8]: # Aggregate service reach by year
reach_by_year = (
    df.groupby("year")["total_clients", "total_undup"]
      .sum()
      .reset_index()
)

reach_by_year
```

Out [8]:

	year	total_clients	total_undup
0	2014	1309195	228769
1	2015	1414290	222409
2	2016	1377194	202866
3	2017	1360974	197427
4	2018	1335373	189803
5	2019	1375411	201614
6	2020	1822792	215916
7	2021	1423014	266640
8	2022	2073753	291815
9	2023	2688188	314680
10	2024	3154795	394582

Aggregate service reach increases over time, with total client counts consistently exceeding unduplicated counts, reflecting repeat utilization within years. Changes in reach reflect both expansion in service coverage and changes in utilization intensity.

```
In [9]: plt.figure()

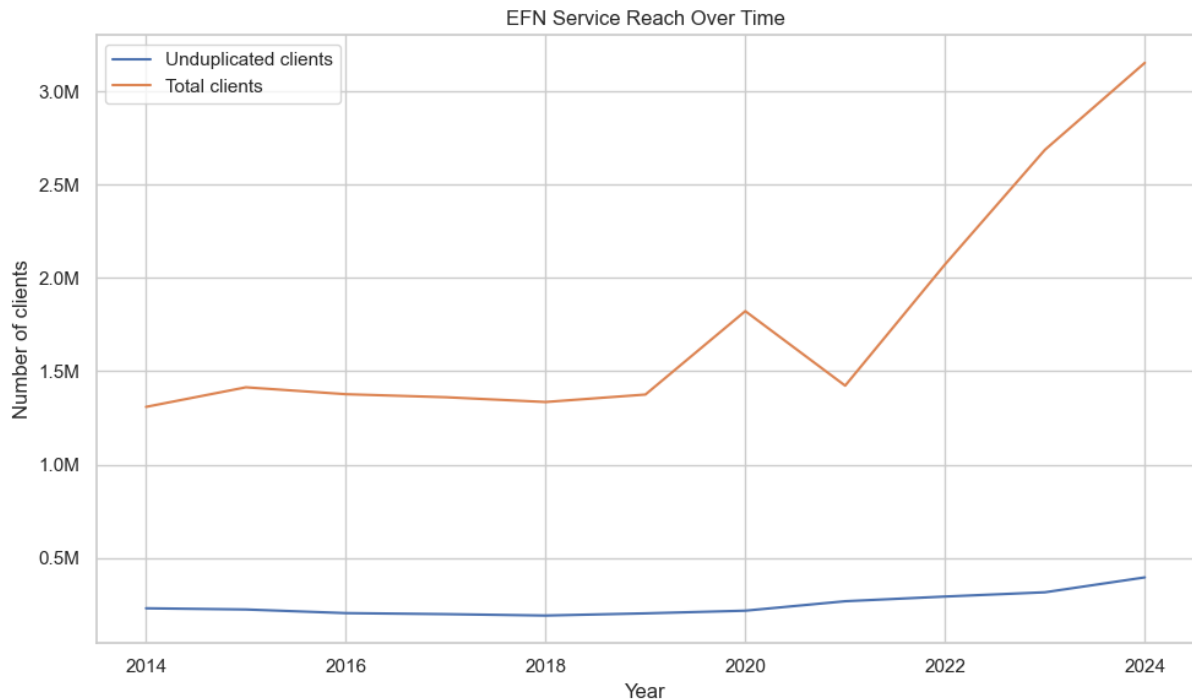
plt.plot(
    reach_by_year["year"],
    reach_by_year["total_undup"],
    label="Unduplicated clients"
)
plt.plot(
    reach_by_year["year"],
    reach_by_year["total_clients"],
    label="Total clients"
)

plt.xlabel("Year")
plt.ylabel("Number of clients")
plt.title("EFN Service Reach Over Time")
plt.legend()

from matplotlib.ticker import FuncFormatter
plt.gca().yaxis.set_major_formatter(
    FuncFormatter(lambda x, _: f"{x/1_000_000:.1f}M")
)

plt.tight_layout()
```

```
plt.show()
```



Service reach exhibits a clear upward trend over the sample period. Growth in total clients outpaces growth in unduplicated clients, indicating increasing repeat utilization alongside expanded reach.

Normalized Service Reach per Service Unit

Interpreting raw service reach trends requires accounting for changes in service unit coverage over time. Observed increases in total or unduplicated clients may reflect either rising demand at existing service units or expansion in the number of active service units. To distinguish between these mechanisms, service reach measures are normalized by the number of active service units in each year.

```
In [10]: # Normalized service reach metrics per service unit
reach_norm = reach_by_year.merge(obs_per_year, on="year", how="left").

reach_norm["undup_per_unit"] = reach_norm["total_undup"] / reach_norm["total_clients_per_unit"]
reach_norm["total_clients_per_unit"] = reach_norm["total_clients"] / reach_norm["num_service_units"]

# Display table (rounded for readability)
reach_norm_display = reach_norm.copy()
reach_norm_display["undup_per_unit"] = reach_norm_display["undup_per_unit"].round(2)
reach_norm_display["total_clients_per_unit"] = reach_norm_display["total_clients_per_unit"].round(2)

reach_norm_display[[
    "year",
    "total_clients",
    "total_undup",
    "num_service_units",
    "undup_per_unit",
    "total_clients_per_unit"
]]
```

```
"undup_per_unit",
"total_clients_per_unit",
]]
```

Out[10]:

	year	total_clients	total_undup	num_service_units	undup_per_unit	total_
0	2014	1309195	228769	29	7889.0	
1	2015	1414290	222409	29	7669.0	
2	2016	1377194	202866	29	6995.0	
3	2017	1360974	197427	30	6581.0	
4	2018	1335373	189803	30	6327.0	
5	2019	1375411	201614	31	6504.0	
6	2020	1822792	215916	31	6965.0	
7	2021	1423014	266640	33	8080.0	
8	2022	2073753	291815	36	8106.0	
9	2023	2688188	314680	36	8741.0	
10	2024	3154795	394582	34	11605.0	

Normalizing service volume by the number of active service units reveals substantial growth in both reach and utilization intensity over time. While the number of service units increases modestly, both unduplicated clients per unit and total service encounters per unit rise sharply after 2020. This pattern indicates that observed growth in total service volume reflects increased demand and repeat utilization at existing service units rather than expansion alone.

3. Service Intensity (Repeat Utilization)

While service reach captures how many individuals are served, it does not describe how frequently clients rely on services within a year. To quantify repeat utilization, this section introduces a measure of service intensity defined as the average number of service encounters per client-year.

Metric Definition

Service intensity is defined as:

$$\text{AvgVisits}_t = \frac{\text{TotalClients}_t}{\text{UnduplicatedClients}_t}$$

where:

- `total_clients` counts all service encounters within a year,
- `total_undup` counts unique individuals served within that year.

Construction Checks

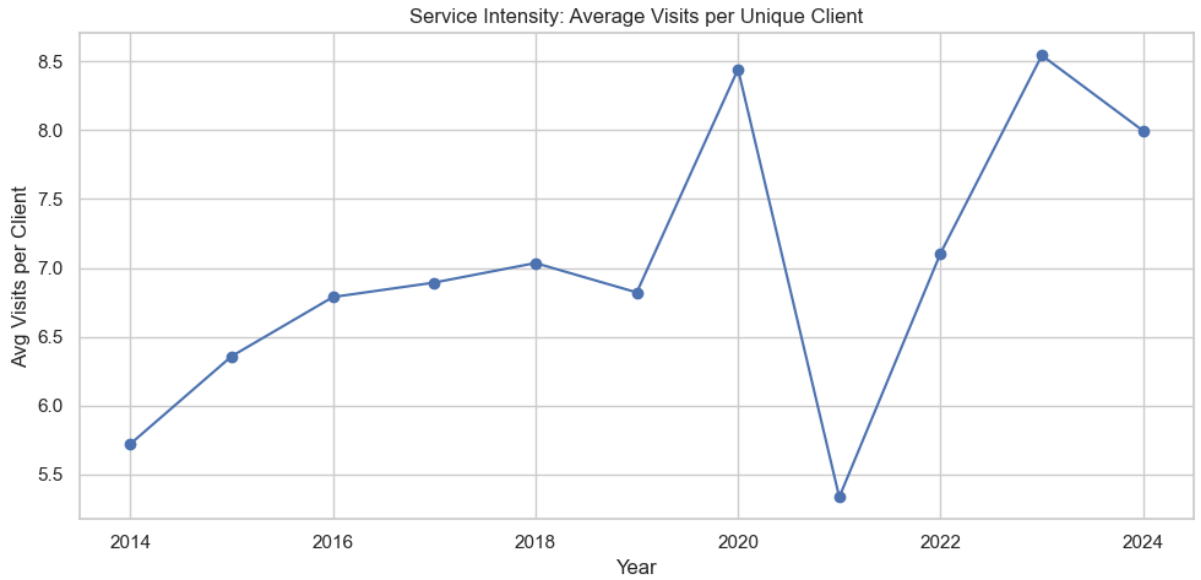
Before analysis, the following properties were verified:

- The denominator (`total_undup`) is strictly positive in all years.
- The metric is consistent with the accounting identity:
$$\text{total_clients} = \text{total_undup} + \text{total_dup}.$$
- No implausible spikes or outliers are observed.

```
In [11]: df_year = (
    df.groupby("year", as_index=False)
        .agg(
            total_clients=("total_clients", "sum"),
            total_undup=("total_undup", "sum"),
            total_dup=("total_dup", "sum"),
            efn_pounds_to_partner=("efn_pounds_to_partner", "sum"),
            pounds_dist_partner=("pounds_dist_partner", "sum"),
            num_service_units=("zip", "nunique"),
        )
        .sort_values("year")
    )
df_year = df_year.copy()

df_year["avg_visits_per_client"] = (
    df_year["total_clients"] / df_year["total_undup"]
)
plt.figure(figsize=(10, 5))
plt.plot(
    df_year["year"],
    df_year["avg_visits_per_client"],
    marker="o"
)

plt.title("Service Intensity: Average Visits per Unique Client")
plt.xlabel("Year")
plt.ylabel("Avg Visits per Client")
plt.tight_layout()
plt.show()
```

Average service intensity increases gradually prior to 2020, followed by a sharp rise in 2020 and a pronounced decline in 2021. Service intensity then rebounds in subsequent years, reaching levels comparable to or exceeding those observed in 2020. These patterns indicate variation in repeat utilization over time and are descriptive in nature.

4. Food Distribution Capacity

Having documented growth in both service reach and repeat utilization, this section examines whether food distribution capacity scaled alongside increased demand. Capacity is evaluated using normalized measures that abstract from changes in network size and client counts.

Metric Definitions

Food distribution capacity is summarized using the following normalized measures:

$$\text{PoundsPerUnit}_t = \frac{\text{PartnerPounds}_t}{\text{ServiceUnits}_t}$$

$$\text{PoundsPerClient}_t = \frac{\text{PartnerPounds}_t}{\text{UnduplicatedClients}_t}$$

Construction Checks

Before analysis, the following properties were verified:

- The denominators used in capacity measures (`num_service_units`, `total_undup`) are strictly positive in all years.

- Aggregated food distribution measures are non-negative in all years.
- Capacity measures are constructed using year-level totals consistent with previously defined service reach and intensity metrics.
- Partner-level food distribution measures (`efn_pounds_to_partner`) are not populated prior to 2022; capacity metrics are therefore interpreted for the post-2021 period.

```
In [12]: df_year = df_year.copy()

df_year["pounds_per_unit"] = (
    df_year["efn_pounds_to_partner"] / df_year["num_service_units"]
)

df_year["pounds_per_undup"] = (
    df_year["efn_pounds_to_partner"] / df_year["total_undup"]
)

df_year[["year", "pounds_per_unit", "pounds_per_undup"]].round(1)
```

```
Out[12]:
```

	year	pounds_per_unit	pounds_per_undup
0	2014	0.0	0.0
1	2015	0.0	0.0
2	2016	0.0	0.0
3	2017	0.0	0.0
4	2018	0.0	0.0
5	2019	0.0	0.0
6	2020	0.0	0.0
7	2021	0.0	0.0
8	2022	13967.5	1.7
9	2023	379618.4	42.2
10	2024	389175.8	32.5

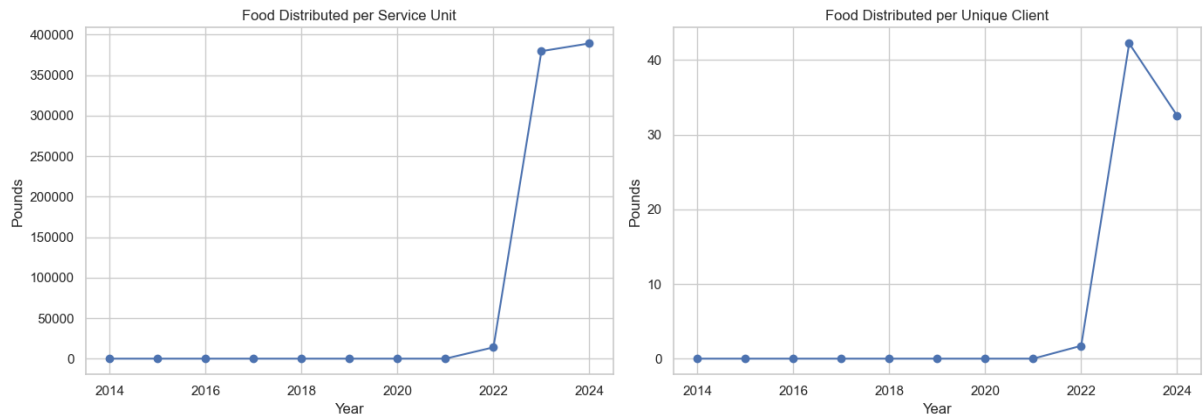
```
In [13]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].plot(df_year["year"], df_year["pounds_per_unit"], marker="o")
axes[0].set_title("Food Distributed per Service Unit")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Pounds")

axes[1].plot(df_year["year"], df_year["pounds_per_undup"], marker="o")
axes[1].set_title("Food Distributed per Unique Client")
```

```
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Pounds")

plt.tight_layout()
plt.show()
```



Food distribution capacity increases substantially in the post-2021 period, coinciding with the first years in which partner-level food distribution measures are observed. Normalization by service units indicates increased throughput at existing locations. When normalized by unduplicated clients, capacity growth broadly tracks increases in service utilization, suggesting that expanded food distribution primarily coincided with higher demand rather than generating excess per-client supply.

These results are descriptive and do not imply sufficiency, efficiency, or constraint.

5. Age-Group Composition and Utilization

Construction Checks

Before analysis, the following properties were verified:

- Age-group counts sum to total counts in each year.
- Age-group unduplicated counts are strictly positive in all years where the group is observed.
- Age-group metrics are constructed using the same year-level aggregation as prior sections.

Age Group Composition Over Time

```
In [14]: # Age-group column lists
undup_cols = ["0-2_undup", "3-18_undup", "19-54_undup", "55plus_undup"]
dup_cols    = ["0-2_dup", "3-18_dup", "19-54_dup", "55plus_dup",
```

```

# Friendly labels for plots
labels = {
    "0-2_undup": "Ages 0-2",
    "3-18_undup": "Ages 3-18",
    "19-54_undup": "Ages 19-54",
    "55plus_undup": "Ages 55+",
    "unknown_undup": "Unknown",
}

# Aggregate unduplicated counts to year level
age_comp = (
    df.groupby("year", as_index=False)[undup_cols]
      .sum()
      .sort_values("year")
)

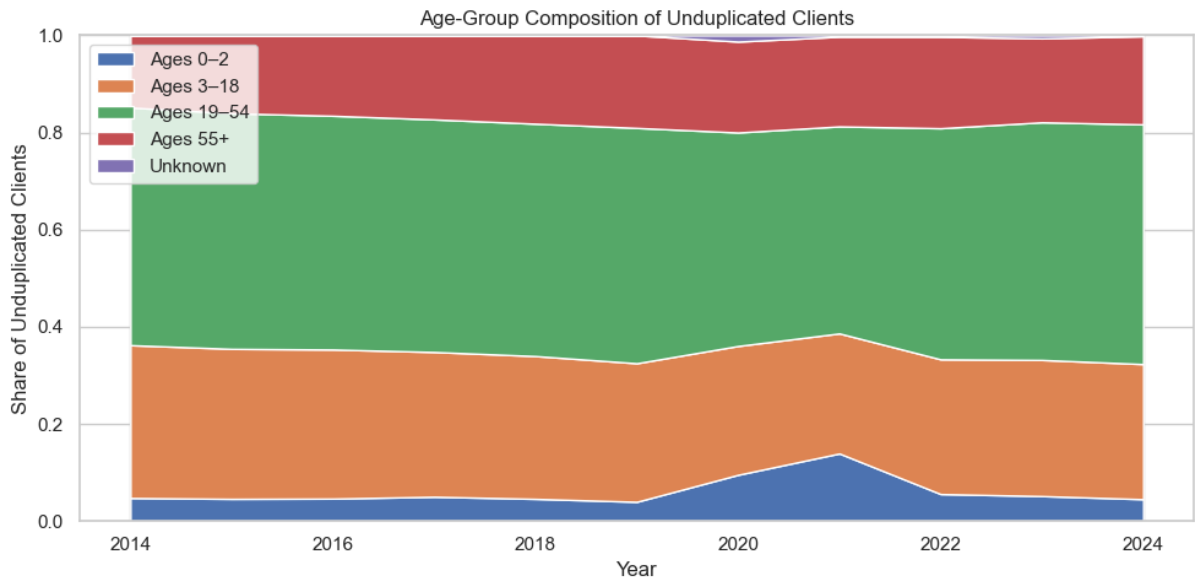
age_comp["total_undup_from_agecols"] = age_comp[undup_cols].sum(axis=1)

# Share of unduplicated clients by age group
for c in undup_cols:
    age_comp[c.replace("_undup", "_share")] = age_comp[c] / age_comp["total_undup_from_agecols"]

share_cols = [c.replace("_undup", "_share") for c in undup_cols]

# Plot: stacked area of composition shares
plt.figure(figsize=(10, 5))
plt.stackplot(
    age_comp["year"],
    [age_comp[c] for c in share_cols],
    labels=[labels[c.replace("_share", "_undup")] for c in share_cols]
)
plt.title("Age-Group Composition of Unduplicated Clients")
plt.xlabel("Year")
plt.ylabel("Share of Unduplicated Clients")
plt.ylim(0, 1)
plt.legend(loc="upper left")
plt.tight_layout()
plt.show()

```



This plot summarizes how the composition of unduplicated clients varies across age groups over time. Shares are computed using the year-level sum of age-group unduplicated counts. The "Unknown" category reflects records that could not be assigned to a specific age group and should be interpreted as a data classification component rather than a demographic group.

Repeat Utilization by Age Group

```
In [15]: # Aggregate age-group undup + dup to year level
age_int = (
    df.groupby("year", as_index=False)[undup_cols + dup_cols]
      .sum()
      .sort_values("year")
)

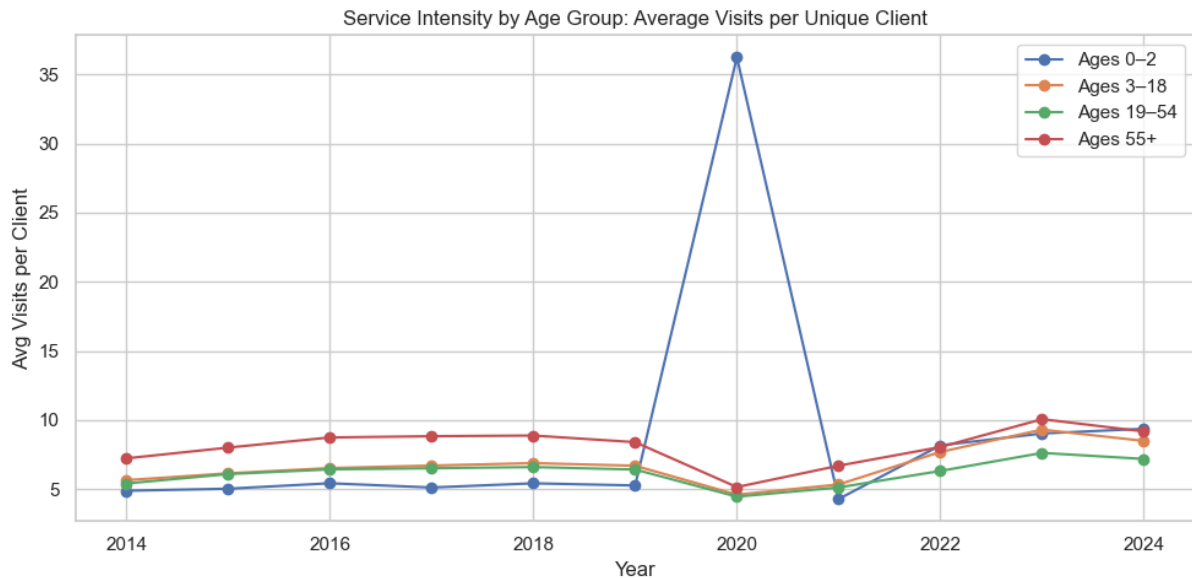
# Helper to compute avg visits safely
def avg_visits(undup, dup):
    return np.where(undup > 0, (undup + dup) / undup, np.nan)

# Compute avg visits per unduplicated client by age group
age_groups = ["0-2", "3-18", "19-54", "55plus", "unknown"]

for g in age_groups:
    u = f"{g}_undup"
    d = f"{g}_dup"
    age_int[f"{g}_avg_visits"] = avg_visits(age_int[u], age_int[d])

# Plot: lines for avg visits by age group
plt.figure(figsize=(10, 5))
plt.plot(age_int["year"], age_int["0-2_avg_visits"], marker="o", label="0-2")
plt.plot(age_int["year"], age_int["3-18_avg_visits"], marker="o", label="3-18")
plt.plot(age_int["year"], age_int["19-54_avg_visits"], marker="o", label="19-54")
plt.plot(age_int["year"], age_int["55plus_avg_visits"], marker="o", label="55plus")
plt.plot(age_int["year"], age_int["unknown_avg_visits"], marker="o", label="unknown")
```

```
plt.title("Service Intensity by Age Group: Average Visits per Unique C
plt.xlabel("Year")
plt.ylabel("Avg Visits per Client")
plt.tight_layout()
plt.legend()
plt.show()
```



This figure compares repeat utilization across age groups using age-specific service intensity measures. Across most years, average visits per unduplicated client are relatively stable within age groups, with older adults (ages 55+) exhibiting consistently higher utilization and adults ages 19–54 comprising a stable middle range.

A pronounced spike in service intensity is observed in 2020 for the youngest age group (ages 0–2). This increase is present in the underlying data and reflects a sharp rise in average encounters per unduplicated client during that year. The magnitude and timing of the spike suggest a period-specific disruption rather than a persistent shift in utilization behavior. This pattern is consistent with temporary changes observed during 2020 and may reflect changes in service delivery, eligibility rules, reporting practices, or household-level consolidation of assistance during the pandemic period.

Following 2020, service intensity for ages 0–2 returns to levels comparable to adjacent years, while other age groups display modest increases in the post-2021 period. Differences across age groups are descriptive and highlight heterogeneity in utilization intensity, but do not imply causal mechanisms or program effectiveness.

6. Variation Across Service Units

To characterize heterogeneity across service units, we first construct a service-unit-level summary by aggregating the ZIP-year panel to the unit level. To avoid conflating scale with panel completeness, totals are converted to per-year averages by dividing by the number of years each service unit is observed.

Construction Checks

Before analysis, the following properties were verified:

- Each service unit is observed for at least one year in the sample.
- Unduplicated client counts are strictly positive for all service units.
- Total service encounters weakly exceed unduplicated client counts for all units, consistent with the accounting identity.
- Service-unit-level totals are normalized by the number of years observed to ensure comparability across units with differing panel lengths.

All results in this section are descriptive and do not imply performance or efficiency differences across partners.

```
In [16]: # 6.0 Service-unit level summary

unit_summary = (
    df.groupby("zip", as_index=False)
      .agg(
        total_undup=("total_undup", "sum"),
        total_clients=("total_clients", "sum"),
        total_dup=("total_dup", "sum"),
        years_observed=("year", "nunique"),
      )
)

# Normalize totals by years observed
unit_summary["avg_undup_per_year"] = (
    unit_summary["total_undup"] / unit_summary["years_observed"]
)

unit_summary["avg_clients_per_year"] = (
    unit_summary["total_clients"] / unit_summary["years_observed"]
)

unit_summary["avg_dup_per_year"] = (
    unit_summary["total_dup"] / unit_summary["years_observed"]
)

# Unit-level service intensity
unit_summary["avg_visits_per_client"] = (
    unit_summary["total_clients"] / unit_summary["total_undup"]
)
```

```
# Display-ready version
unit_summary_display = unit_summary.copy()
unit_summary_display[[
    "avg_undup_per_year",
    "avg_clients_per_year",
    "avg_dup_per_year",
    "avg_visits_per_client"
]] = unit_summary_display[[
    "avg_undup_per_year",
    "avg_clients_per_year",
    "avg_dup_per_year",
    "avg_visits_per_client"
]]

unit_summary.head().round(2)
```

```
Out[16]:
```

	zip	total_undup	total_clients	total_dup	years_observed	avg_undup_pe
0	98321	11568	84476	72908	11	1
1	98328	27273	231408	204135	11	2
2	98329	14506	94631	80125	11	'
3	98335	31061	133630	102569	11	2
4	98338	19947	177071	157124	11	1

The table above summarizes service-unit-level utilization by aggregating ZIP-year observations to the unit level and normalizing totals by the number of years each unit is observed. Reported values therefore reflect average annual activity rather than cumulative scale.

The unit-level service intensity measure (avg_visits_per_client) captures the average number of service encounters per unduplicated client within each service unit across the observed period. Differences across units reflect heterogeneity in utilization patterns and program structure and are descriptive in nature.

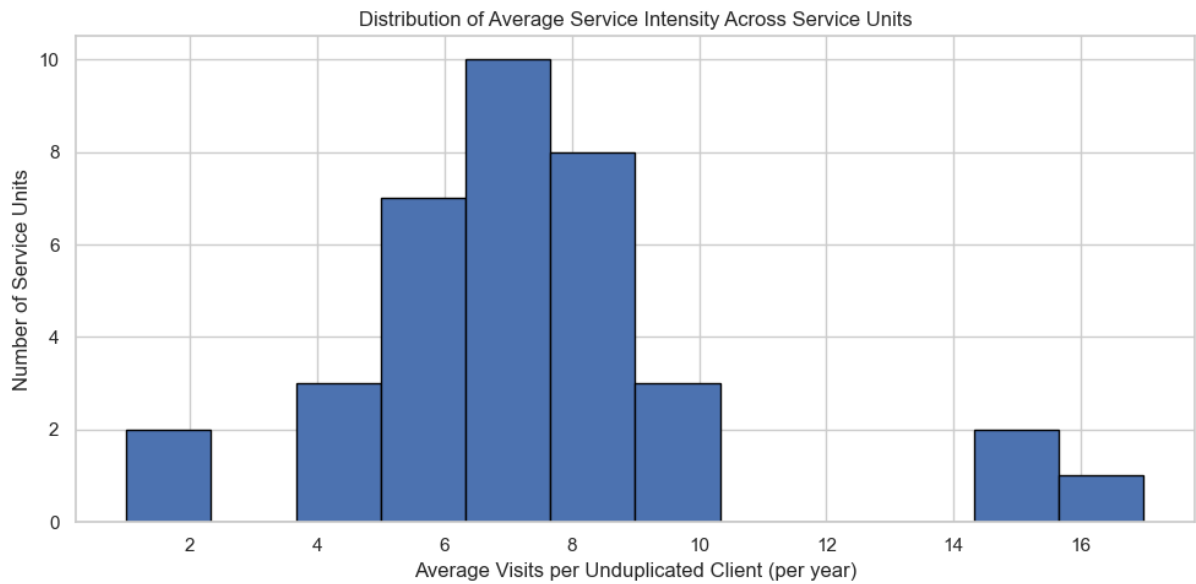
Distribution of Service Intensity Across Service Units

```
In [17]: plt.figure(figsize=(10, 5))
plt.hist(
    unit_summary_display["avg_visits_per_client"],
    bins=12,
    edgecolor="black"
)

plt.title("Distribution of Average Service Intensity Across Service Un
plt.xlabel("Average Visits per Unduplicated Client (per year)")
plt.ylabel("Number of Service Units")
```



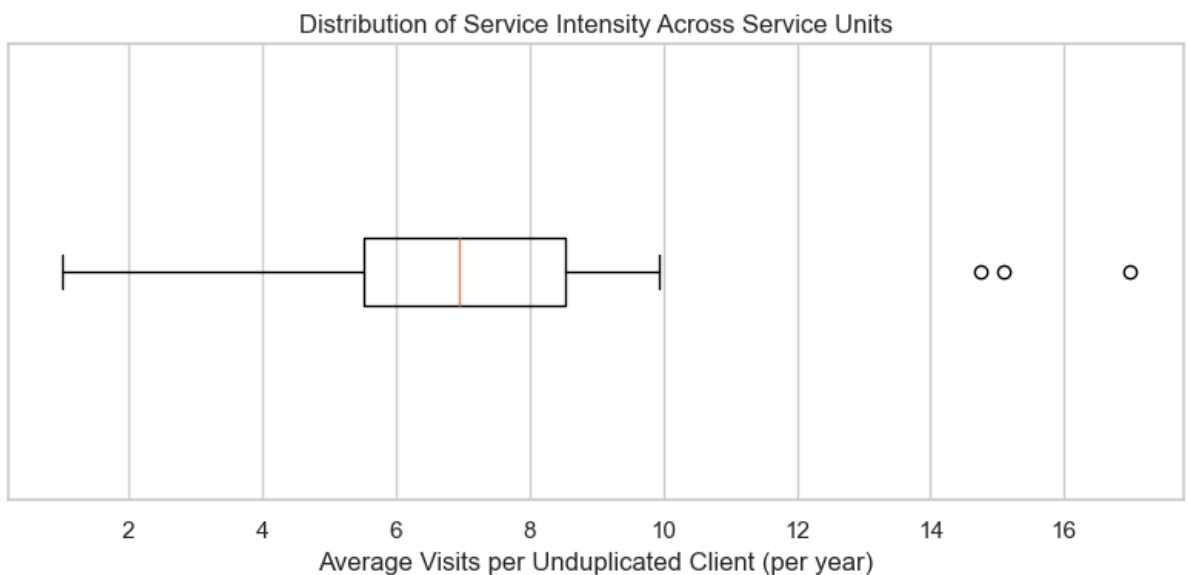
```
plt.tight_layout()
plt.show()
```



```
In [18]: plt.figure(figsize=(8, 4))
plt.boxplot(
    unit_summary_display["avg_visits_per_client"],
    vert=False
)

plt.yticks([]) # hide dummy y-axis
plt.title("Distribution of Service Intensity Across Service Units")
plt.xlabel("Average Visits per Unduplicated Client (per year)")

plt.tight_layout()
plt.show()
```



These figures summarize the distribution of service intensity across service units,

measured as the average number of service encounters per unduplicated client per year. Service units exhibit substantial heterogeneity in utilization intensity, with most units clustered within a moderate range and a smaller number of units displaying notably higher average visit counts. These differences are descriptive and reflect variation in service models, client populations, and program structure rather than performance or efficiency.

7. Summary of Descriptive Patterns

This exploratory analysis documents several consistent patterns in Emergency Food Network service utilization over the 2014–2024 period.

First, overall service reach increased substantially over time, particularly after 2020. Growth in total service encounters exceeded growth in unduplicated clients, indicating increased repeat utilization rather than expansion driven solely by the number of individuals served. Normalization by the number of active service units shows that this increase reflects higher throughput at existing locations in addition to changes in network size.

Second, service intensity—measured as the average number of encounters per unduplicated client—remained relatively stable in the pre-2020 period, increased sharply during 2020, and subsequently returned toward levels comparable to adjacent years. This pattern is consistent with a temporary disruption during the pandemic period rather than a persistent shift in utilization behavior.

Third, food distribution capacity increased markedly in the post-2021 period, coinciding with the first years in which partner-level food distribution measures are observed. When normalized by service units, capacity growth indicates increased throughput at existing locations. When normalized by unduplicated clients, capacity growth broadly tracks increases in service utilization, suggesting that expanded food distribution coincided with higher demand rather than generating excess per-client supply.

Fourth, demographic composition of unduplicated clients remained relatively stable across age groups over time, with adults ages 19–54 comprising the largest share of clients throughout the sample. Service intensity differed modestly across age groups, with older cohorts exhibiting higher average utilization and a pronounced, temporary spike observed for the youngest age group (0–2) during 2020. These differences are descriptive and reflect heterogeneity in service utilization patterns rather than causal mechanisms.

Finally, substantial variation exists across service units in average service

intensity. Most units cluster within a moderate range of utilization, while a smaller number exhibit notably higher average visit counts. This heterogeneity reflects differences in service models, client populations, and program structure and does not imply performance, efficiency, or effectiveness differences across partners.

Overall, the results presented here are descriptive in nature and provide a structured overview of temporal trends, utilization intensity, demographic composition, and service-unit heterogeneity within the EFN service network. These findings motivate, but do not substitute for, more targeted analysis of underlying mechanisms, policy changes, or program-level effects.